



COMPONENTS

Low-power MCUs

STM32WB35 and STM32WB30 Value Line are the new dual-core multi-protocol MCUs from STMicroelectronics for large-scale IoT applications



such as fleet and resource management, as well as asset monitoring and tracking. The

extremely low-power MCUs integrate a 2.4GHz radio managed by a dedicated Arm Cortex-M0+ core, with a 64MHz Arm Cortex-M4 for the main application for permitting uninterrupted real-time performance. They support Bluetooth Low Energy 5.0, Zigbee 3.0, OpenThread (IEEE 802.15.4) protocols and concurrent modes, with built-in security features. The devices include up to 512kB integrated Flash memory and up to 96kB SRAM, with a Quad-SPI interface for connecting to external high-speed memory. They also feature secure firmware installation, hardware encryption for the application and radio stacks, hardware public-key authority, hardware accelerators for state-of-the-art cryptographic algorithms, and support for secure OTA firmware updates.

STMicroelectronics
www.st.com

Motor controller

The LX7720 radiation-hardened mixed-signal motor controller from Microchip aims to reduce weight and board



space. By reducing the number of components on a system, developers can inspect and test fewer parts, while also minimising the physical points of potential failure

due to a smaller number of connections and solder joints. Some of the integrations into this device are: four half-bridge N-channel MOSFET

drivers, four floating differential current sensors, pulse modulated resolver transformer driver, three differential resolver sense inputs, six bi-level logic inputs, power drivers via external FETs, loop control electronics for voltage or current control, and position read-back. It serves as a mixed-signal companion IC to the selected digital IC used in the application and is ideal for enabling various motor control applications in space robotics and human-rated space missions.

Microchip Technology
www.microchip.com

Diode array

SP1250 discrete unidirectional TVS diode array from Littelfuse provides high electrostatic discharge (ESD) protection for consumer



electronic equipment by safely absorbing repetitive ESD strikes of $\pm 30\text{kV}$ (contact and air discharge) without any performance degradation. Each device

can dissipate up to 50A, 8/20 μs surge event. Typical applications for the SP1250-01ETG include desktops, servers, notebooks, computer peripherals, medical equipment, POS terminals, test equipment, and portable battery.

Littelfuse
www.littelfuse.com

Buck boost DC-DC converters

The ISL9122A, a flexible buck-boost switching regulator with bypass mode from Renesas, provides ultra-low quiescent current (I_Q) < 1.3 μA for powering sensors, MCUs, wireless devices, and



other system components. It features low shut-down mode current of

7nA to extend battery life, high efficiency at light load (84% at 10 μA) and 97% peak efficiency to reduce power drain and heat buildup, automatic and selectable forced bypass power-saving mode to reduce I_Q , output current up to 500mA ($V_{out} = 3.3\text{V}$, $V_{in} = 3.6\text{V}$), ultra-small footprint with 1.8 $\times$ 1.0mm WLCSP package or 3 $\times$ 2mm DFN package, requires only three external components (inductor, input and output capacitors), full protection against over-current, short-circuit and overheating. Operating off 1.8V to 5.5V, it extends the battery life of smart IoT devices. Target applications include wireless earbuds, fitness bands, smartwatches, water and gas meters, portable medical devices and battery-operated smart IoT devices.

Renesas Electronics Corporation
www.renesas.com

DC-DC converter in SOIC package

R05CT05S, a high-isolation, high power density, 0.5W DC-DC converter from RECOM provides a fully regulated, low-noise output. It has a nominal 5V input (4.5 – 5.5V) and is settable



by a resistor to 3.3V or 5V DC for direct connection to the load, or

3.7V/5.4V DC to power a downstream LDO for ultra-low noise applications. Protection and control features include short-circuit immunity, over-temperature shutdown, and under-voltage lockout. It is suitable for medical applications up to 5000m altitude and applications with high common-mode, switched voltages such as high-side gate driver power supplies.

RECOM
www.recom-power.com

Graphic controller ICs

To facilitate larger displays at greater resolution, graphic controller ICs BT817 and BT818 have been introduced by